(i) $N = 3^2 \cdot 5^3$

$$\phi(N) = (2+1)(3+1) = 12$$

Excluding 1 and N, there are $12 - 2 = 10$ proper factors.

$$(m+1)(n+1) - 2 = 10 \implies (m+1)(n+1) = 12$$

$$(m+1, n+1) = (1,12), (2,6), (3,4), (4,3), (6,2), (12,1)$$
$$\implies (m,n) = (0,11), (1,5), (2,3), (3,2), (5,1), (11,0)$$

Removing $(2,3)$, $\boxed{5}$ combinations

(ii) Let $\phi(N) = 428 = 4 \cdot 107 = 2 \cdot 2 \cdot 107$
For the smallest N, put larger exponents on smaller primes.
Possible exponent sets: $427 \quad 213, 1 \quad 106, 3 \quad 106, 1, 1$
Correspondingly: $2^{427} \quad 2^{213}3 \quad 2^{106}3^3 \quad 2^{106}3 \cdot 5$

$$\therefore \boxed{N = 2^{106} \cdot 3 \cdot 5}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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